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% Yeshwantrao Chavan College of Engineering, Nagpur
% Department of Electronics and Telecommunication Engineering
% Session ODD 2024-25
% Name of Student: Aditya Bhongade
% Sec: 5-ET-A
% Roll No: 43
% Name of Course Teacher : Dr. Y. U. Chitriv(Dubey)/Prof. Y. S. Kale
```

Experiment 7 : High Pass Filtering (Sharpening) in Spatial Domain

```
clc; close all; clear;
```

Unsharp Masking & High Boost Filtering

```
I = im2double(imread('Images3/dip_xe.tif'));

f = im2double([0 0 0; 0 1 0; 0 0 0]);
f_bar = im2double(ones(3, 3) / 9);

gm = f - f_bar;

r = f + gm;
r1 = f + 5.*gm;

g = imfilter(I, r, 'same');
g1 = imfilter(I, r1, 'same');

figure,
subplot(2, 2, [1 2]), imshow(I), title("Original Image");
subplot(2, 2, 3), imshow(g), title({"Unsharp Masking"});
subplot(2, 2, 4), imshow(g1), title("High Boost Filtering");
```

Original Image



Unsharp Masking



High Boost Filtering



Conclusion

The input images were effectively sharpened using Unsharp Masking and High Boost Filtering. Both techniques produced output images that exhibited improved detail compared to the originals. Among them, High Boost Filtering ($K = 5$) delivered superior enhancement and more pronounced edge highlighting than Unsharp Masking ($K = 1$).